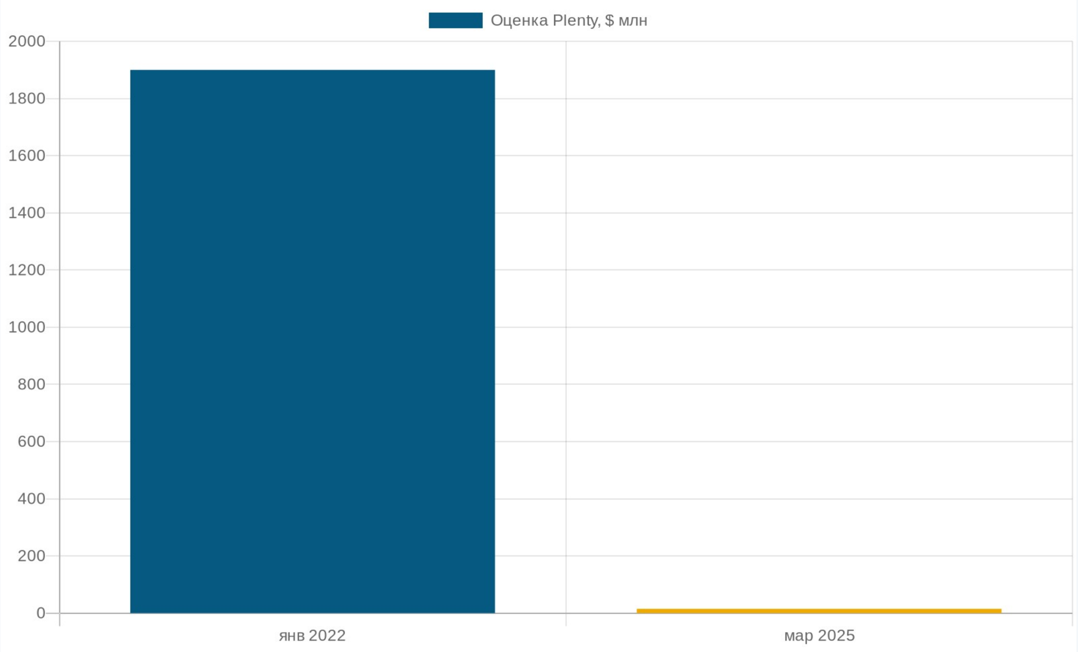
An AI-run farm promised a revolution. 19 months later — it shut down.

May 2023 · Plenty Compton opening



December 2024 · shutdown · Ch. 11 March 2025

$940M
capital lost

$1.9B → <$15M
valuation collapse –99% in 3 years

19 months
from opening to shutdown

Sources: Plenty press May 2023; TechCrunch 2025-03-24; Bloomberg Law 2025

10

Lecture 10 · AI in Agriculture · Year 3

AI in Agriculture

From field to shelf: the five-level ladder and five anti-AI criteria

Q&A at the end

Lecture map — the five-level ladder from field to shelf

0. Opening	1. L1 Field	2. L2 Robot	3. L3 Animal	4. L4 Chain	4-bis. Environment	5. L5 Shelf
Hook + axis + glossary	*Open biology*	*Specialization > generality*	*Semi-closed environment, individual*	*Agentic AI leads*	*Connectivity · lock-in · regulation*	*+ 5 criteria + closing*
	5 cases · 3 failures	***4 cases · 3 failures***	***4 cases · 3 lessons***	***4 cases · 2 failures***	***3 sub-blocks***	***L5 + 5 AP + checklist***

Central question of the lecture:

Where on the L1→L5 ladder does AI work, where does it break — and which class of solution / alternative applies at each rung?

Two AI modes and five L4 terms — the shared language of the lecture

AI operating environment

Closed-loop

Controlled environment (greenhouse, factory, warehouse). Every variable is measured. AI optimizes parameters. Production deployment is mature.

Open environment

Open biological environment (field, meadow). Weather, variable lighting, dust, shadows. AI works in narrow tasks. The generic "autonomous farm" goes bankrupt.

5 supply-chain terms

Agentic AI	*ML + retrieval (RAG) + action pipeline + human-in-the-loop*
bp (basis point)	*1 bp = 0.01%. A metric of trade slippage*
Slippage	*The gap between the planned trade and its actual execution*
Scope-3 emissions	*Indirect emissions across the supply chain*
AI-MRV	*AI monitoring, reporting, verification — climate accounting*

The AI ladder — five levels. Each rung works differently

L5	**Shelf / store**	*Fully digitized environment. AI is mature.*	*Walmart Eden 2017+, Tesco 2017+, X5 2020+*
L4	**Supply chain**	*Controlled goods flows + fast feedback*	*Cargill CMAX, Tract, Olam, Walmart×Cropin*
L3	**Animal**	*Semi-closed environment + individual-level measurement*	*SenseHub 2M cows, CattleEye, DeLaval V310*
L2	**Robot**	*Specialization works; the generic one goes bankrupt*	*LaserWeeder G2, Saga UV-C, Tevel · vs Monarch/FarmWise*
L1	**Field**	*Open biology — the hardest*	*See & Spray works · Plenty/AppHarvest/Bowery failed*

↑ moving up the ladder

↑ controllability
of environment rises

↑ measurable return
ROI rises

↑ predictability
of outcome rises

↓ biology
uncertainty falls

↑ environment controllability ↔ ↑ AI penetration · next — L11 cyber-physical manufacturing

1

Section 1 — L1 "Field"

Open biological environment: where AI works narrowly — and where it breaks even here

1 working case + 3 vendor matrix · 3 failures (vertical farming, ChatGPT, Plantix) · 2 anti-AI criteria

See & Spray Ultimate — the canonical L1 success

John Deere ExactApply + See & Spray Ultimate · November 2025

5M acres

2025 season

≈0.55% of 900M-acre US ag total

–50%

contact herbicides

from baseline ≈1 lb/acre AI → ≈0.5 lb

+2.0 bushels

soybeans per acre

from US avg ≈177 bu/A = +1.1%

36 cameras · CNN on millions of images (>1M) · NVIDIA Jetson edge · <50 ms

Sources: AgTechNavigator 2025-11-10; John Deere press release November 2025; GrowIWM 2024

L1 platforms — five vendors, different modes. Brand ≠ operating mode

Platform	Geography	Operating mode	Business model
BASF xarvio	EU, Japan (rice)	Subscription + advisory	Pay per acre
Climate FieldView	US (left Russia 2022)	Data warehouse	250M acres of subscriptions
Syngenta Cropwise	Global	Bayer Forward integration	Bundled in a package
Granular (Corteva)	US	Farm management	Cloud service
Taranis	Global	Computer vision + drones	Pay per acre

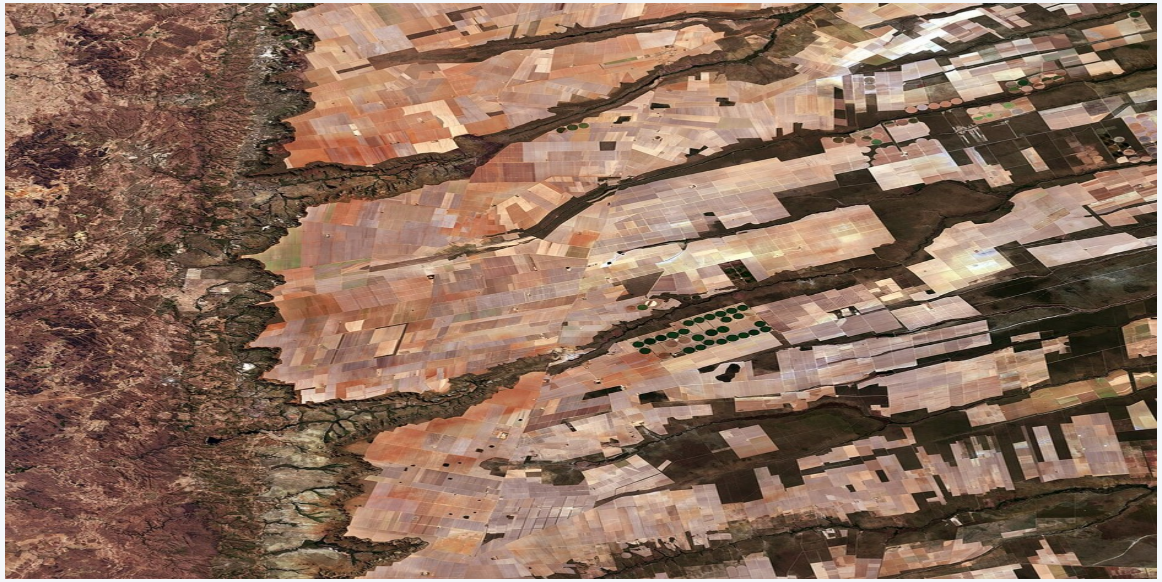
★ Key point on this slide — BASF xarvio in Japan: a rice-yield guarantee (October 2025)

The first case of a guaranteed yield under AI advisory: BASF pays compensation if the harvest falls short. A closed data loop + an insurance mechanism = the only known example as of 2026.

The other 4 platforms in the matrix are for self-study; the "brand ≠ operating mode" criterion applies to each.

Source: BASF press release p-25-191 October 2025 · TAdviser FieldView 2022

Foundation models 2026 — the barrier to entry dropped by 2-3 orders of magnitude



Sentinel-2 / Copernicus (ESA, CC-BY-SA) — the data class for TerraMind

TerraMind (IBM + ESA, 2025)

"GPT-3 for Earth observation". A team of 3 fine-tunes on thousands of images instead of millions.

Vendor concentration risk

The entire L1 industry runs on 2-3 foundation models (IBM / NASA / ESA).

A model shutting down = teams lose capabilities all at once.

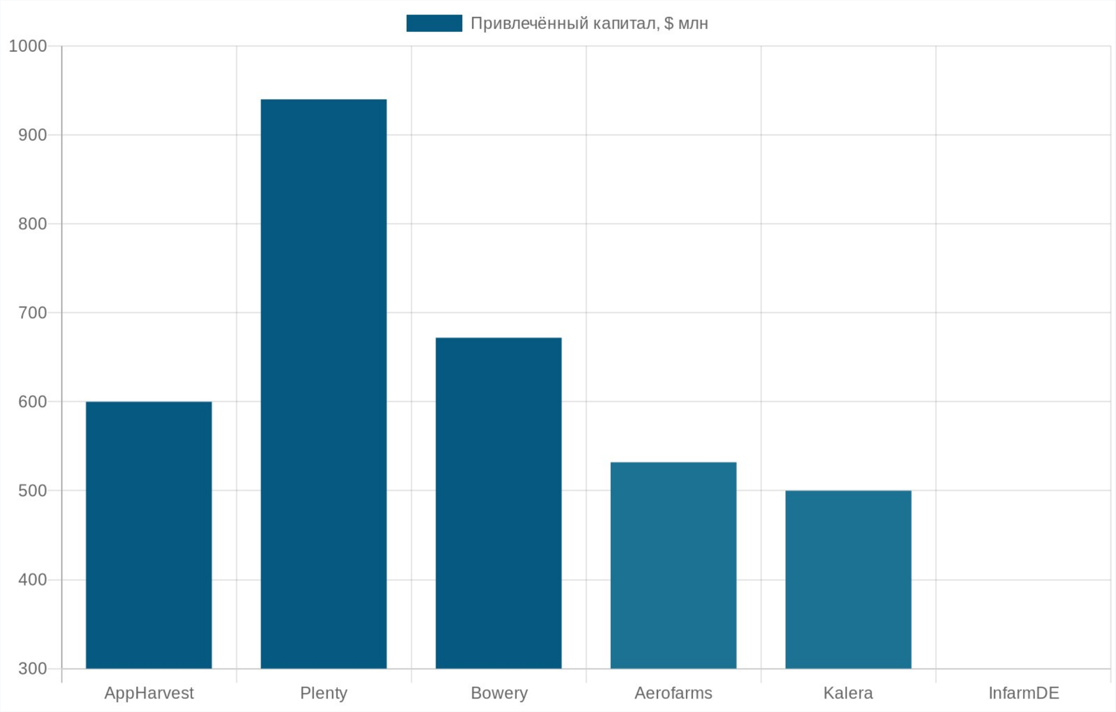
The 2026 advisor architecture

Foundation model (TerraMind)
+ retrieval (RAG) to the local regulator
+ LLM generation

+ explicit abstention at low confidence.

Sources: IBM Research, April 2025; NASA Earth Observatory 2025

Vertical farms — a failure not caused by bad AI. AppHarvest, Plenty, Bowery



AppHarvest

Greenhouse, unprofitable 2023

$475M SPAC + $341M debt ≈ $816M

Plenty

Compton AI factory 19 mo → bankruptcy

$940M of $1B+ raised since 2014; –99% valuation

Bowery

Collapse before IPO, November 2024

>$700M raised; $32M never-used Locust Grove

$1.37B+ in losses, 14 bankruptcies in 2025. The energy arithmetic did not work.

The climate controllers worked. Computer vision recognized. The model predicted. LED ≈ 100× the energy of sunlight — AP1 thermodynamics > AI.

Sources: TechCrunch 2025-03-24, 2024-11-04 (Bowery); Agriculture Dive 689039 (AppHarvest)

Five "whys" — why AI did not close the thermodynamic gap

1. Why did it close?

Capital ran out

2. Why the capital?

Unit economics don't work

3. Why don't they work?

Cost of LED energy 100× sunlight

4. Why 100×?

Law of thermodynamics

5. Can AI close it?

No: effect 5-15% vs a 100× gap

Root cause: a two-order-of-magnitude gap. No model will close it.

0.5 (sun → LED) × 0.7 (LED → plant) × 0.3 (plant → yield) ≈ 10.5% end-to-end. AI optimizes the denominator (5-15%); the gap is in the numerator.

AP1 in its strict form: "when AI optimizes an incorrectly formulated objective function — better not AI".

Analysis by Hannah Ritchie · MDPI Sustainability, 2024

ChatGPT and Bard as agronomists — "confidently wrong" in tens of percent of cases

Tzachor et al., Nature Food, November 2023

Reichman University · 184 questions (publication 11.2023, press 05.2024)

GPT-3.5	**32%**	*correct*
GPT-4	**44%**	*correct*
Bard	**29%**	*correct*

The other 56-71% of answers are confidently wrong: the model does not say "I don't know", it produces a confident answer with the wrong dosage / an unsuitable chemical. The failure mode is not "the model is bad", but "the application is outside its mode".

AP4 — a categorical anti-pattern

A generic LLM in the role of agronomist advisor — categorically inapplicable. Not "needs more work"; this is a different class of task.

Alternative: AI with source verification (RAG)

- A bounded farm knowledge base
- Retrieval + source citation
- Calibrated confidence: "I don't know" when it doesn't
- An audit trail for every recommendation

+ human-in-the-loop: the final recommendation is confirmed by an agronomist

Source: Tzachor et al., Nature Food, November 2023 (press coverage Phys.org 2024-05)

Plantix — 10-15% misdiagnosis × 10M+ downloads = ~100k wrong recommendations / year

Plantix · mobile app

10+ million downloads

[camera]

Photo of a soybean leaf

— analysis —

Diagnosis:

anthracnose

Confidence: 87%

What's wrong with "85-90% accuracy"

- Vendor self-report (no independent audit)
- 10-15% misdiagnosis = 1-1.5M errors / year
- Dose-critical errors (wrong chemical)
- Chronic acceptance of false positives in an advisor

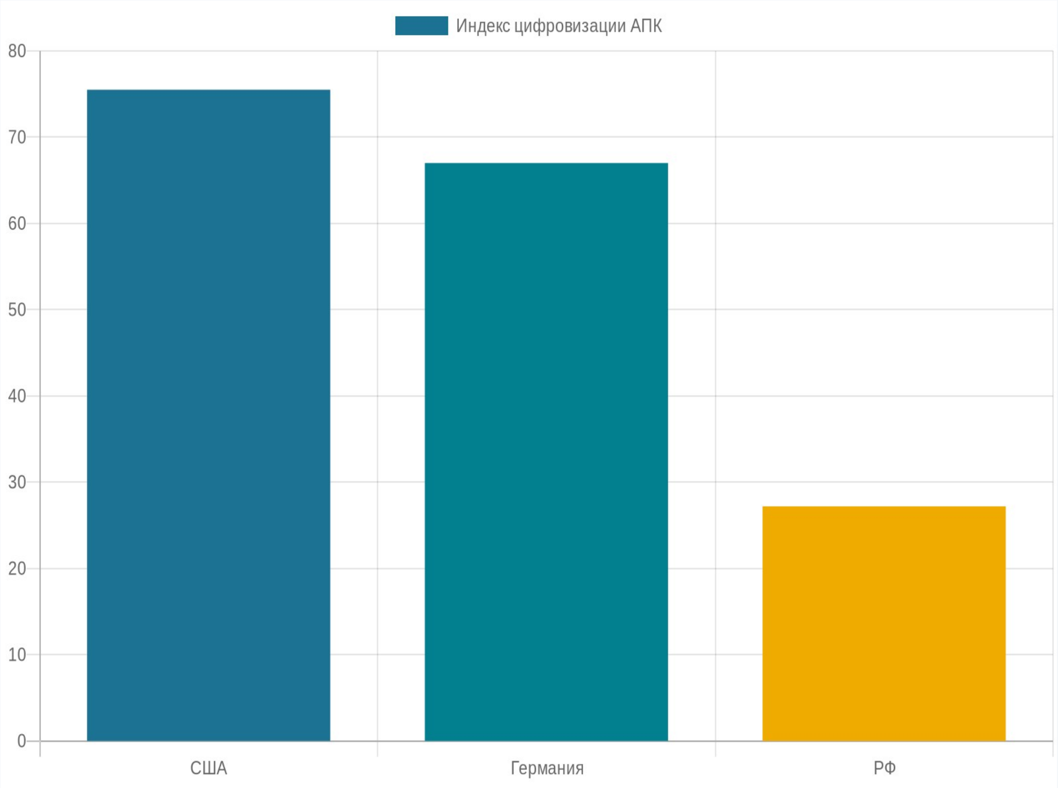
AP3 + alternative

AP3: threshold accuracy ≠ readiness for deployment.
Alternative — calibrated confidence + abstention:

- *output only at ≥90% confidence,*
- *otherwise — "refer to a specialist",*
- *+ partition by dose criticality.*

Plantix.net (interface); data from Frontiers in Plant Science 2020 + Plantix self-report

L1 in Russia — ExactFarming and Progress Agro at a digitalization index of 27.2



ExactFarming

12,700 farms · 9.8M ha

Field management + monitoring

Progress Agro group

+5% profitability

Internal measurement 2024

AP6 — Climate FieldView left Russia in 2022

L1 political risk: cloud services with foreign dependencies can be cut off by a vendor decision or by sanctions.

Sources: Yakov & Partners 2024 · ExactFarming.com · TAdviser 2022

2

Section 2 — L2 “Robot”

Specialization beats generalization. Narrow wins against generic failures.

4 working specialization cases · 3 failures (Monarch, FarmWise, strawberry) · Russia parallel CV vs sensor-fusion

Carbon Robotics LaserWeeder G2 — the canonical L2 success

AI-driven weeding robot (representative photo · "Titan / LaserWeeder" class)

250,000 acres

≈0.028% of 900M acres US ag

15 billion

weeds killed

$1.4M

vs sprayer ~$50k + chemicals $200-400/acre → payback 3-4 years at 1k+ acres

240 W laser

+ CNN on 40 million images

Replacing chemistry with physics — a narrow operation substitution, not an "autonomous tractor." The working L2 pattern.

Source: Carbon Robotics press release, businesswire 2025-02-10

Cognitive Pilot and ITELMA are not AI vs non-AI; they are two different classes of AI

Cognitive Pilot · computer vision

“What I see”

1700+ installations (tractors, combines)

4 farmer lawsuits (~₽12.7M)

Failure mode:

- dust obscures the cameras
- shadow shift changes the lighting
- edge-row recognition breaks down
- benchmark labeling ≠ field conditions
- edge cases not in the training set

ITELMA Kvadro · sensor fusion

Navigation + inertial + corrections: “where I am”

GLONASS + GPS + Galileo + BeiDou

RTK correction: accuracy 2-5 cm

Why it works:

- sensors do not depend on lighting
- multiple systems = redundancy
- GLONASS works in Russia independently
- stable in dust / fog / at night
- compatible with deterministic guidance

Sources: RTVI 2025 (Cognitive Pilot lawsuits); Fontanka 2026-01-26 (ITELMA)

Narrow L2 wins — Solinftec, Saga UV-C, Tevel, AGCO Outrun

Solinftec Solix

+243% year over year in 2025

Vendor self-report: ≤98% reduction in contact herbicides

Saga UV-C

20% of UK strawberries

Ultraviolet nighttime treatment. Not harvesting — disease control

Tevel

Apple-picking drones

Narrow niche — flying drones for picking apples

AGCO PTx Outrun

Retrofit for mixed fleets

AGCO module for a mixed tractor fleet (not AGCO only)

The specialization pattern: every solution is narrow and measurable

⚠ Do not confuse: Saga UV-C ≠ Saga for harvesting. Those 20% of UK strawberries are disease control, not harvesting.

Sources: Solinftec.com · sagarobotics.com · tevel-tech.com · agcocorp.com 2025

Monarch Tractor — “autonomous” with autonomy broken

2018
Founded
Livermore, CA

2022
MK-V demo
Big PR wave

Sept 2025
Burks Tractor lawsuit
10 tractors · $773,088
“unable to operate autonomously”

Nov 2025
102 layoffs
~38% (102 / ~270 after 10% cut Nov 2024)

Apr 2026
Caterpillar acqui-hire
End of independence

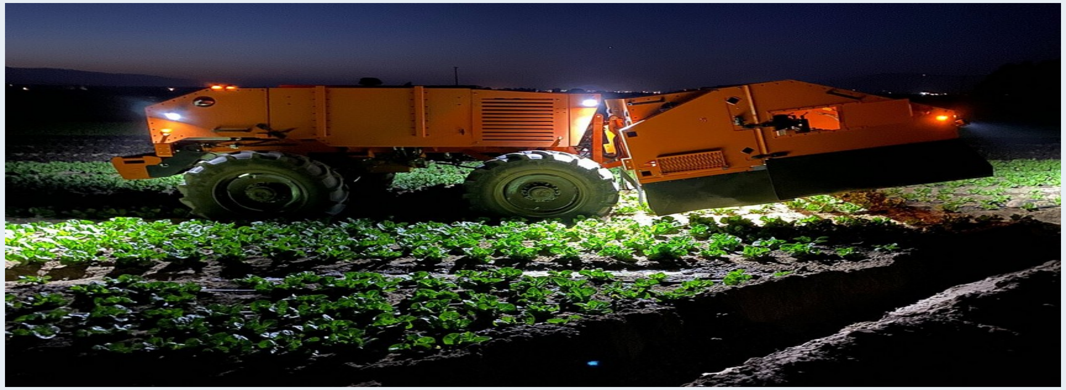
Anti-pattern: demo ≠ production deployment

“Autonomous” in marketing = a legal trap if the actual behavior depends on the operator.
Every autonomy claim must be checked with an independent operator test → if it requires operator presence, the autonomy is partial.

Sources: TechCrunch 2025-11-18, 2025-11-19, 2026-04-15

FarmWise wound down + Naïo –40% — the open environment breaks computer vision

Why the CV stack (computer vision) failed



FarmWise Titan FT35 · wound down in 2025

FarmWise wound down 2025
Naïo €4M → €2.4M (–40%)
Computer vision trained under greenhouse conditions:
• shadow shift (arXiv 2508.19511)
• variable lighting
• dust / fog

AP2b — the deterministic alternative

Lemken / Kverneland — mechanical cultivator

Mechanical cultivators as the deterministic option:
• stem detection by mechanics
• inter-row cultivation
• works in any weather
• cost 3-5× lower than a CV robot

Sources: Farm Progress 2025; arXiv 2508.19511 (shadow bias); Lemken / Kverneland press

Strawberry-picking robot — 10 years of pilots, not production deployment

Manual strawberry harvesting (Wikimedia Commons, CC-BY-SA)

“Harvesting is the last great unsolved problem”

$200-350k — *capital cost per robot*

$68-130k — *annualized depreciation per robot*

$43k / acre — *cost of manual harvesting in California*

<5% of market — *robots × $50B addressable labor market*

Sources: Wikimedia Commons; RobotToday market analysis 2024

3

Section 3 — L3 «Animal»

Semi-closed environment + individual-animal measurement. AI works more reliably than at L1-L2.

4 working cases (SenseHub, CattleEye, DeLaval, Birdoo) · 3 anti-hype lessons (Cainthus, tie-stall, Holstein bias)

0. Opening | 1. L1 Field | 2. L2 Robot | **3. L3 Animal** | 4. L4 Chain | 5. Environment | 6. L5 Shelf

Allflex SenseHub — 2M cows with sensors (≈0.75% of 265M dairy cows worldwide)

Sensor ear tag · Wikimedia (Cow with ear tag, CC-BY-SA)

+ accelerometer collar + cloud analytics

AI pipeline (early warning)

1. Sensor

Accelerometer · 5-7 yr battery

2. Cloud

ML behavior classification

3. Alert to farmer

Early warning, not prescription

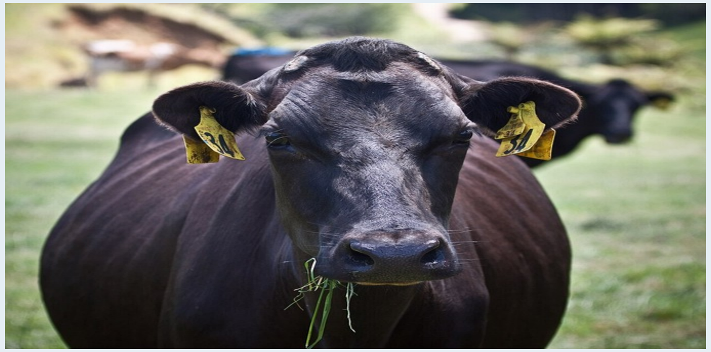
5 alert classes

estrus · calving · lameness · mastitis · pneumonia

AI = augmentation, not replacement

Source: Merck Animal Health, 2025 — 2M dairy-cow milestone

CattleEye + DeLaval VMS V310 + Cargill Birdoo — 3 working cases



CattleEye

60 farms · 11,000 cows

CCTV + cloud AI:
lameness scoring · GEA network >250,000

DeLaval VMS V310

99.8% attachment rate

998/1000 successful attachments
of the milking-unit arm

Cargill Birdoo

>95% accuracy

Computer vision for weight
estimation · saves 10-30 g feed / broiler

Sources: Fortune 2025-06 (CattleEye GEA); DeLaval press 2025-04; Cargill.com 2025

Cainthus, tie-stall barns, Holstein bias — 3 anti-hype lessons of L3

Cainthus (Cargill 2018)

Announcement ≠ deployment

No public production metrics. A branded press release, but 0 measurable deployment data by 2026.

Tie-stall barns

Barn architecture breaks computer vision

The algorithm works in free-stall (loose) housing, not in tie-stall.

Most Russian cows are tie-stall housed.

Holstein bias

Training data ≠ local breeds

AI trained on Holstein / Friesian breeds. Yaroslavl, Yakut, Kholmogory — different morphology.

AI capability ≠ AI applicability.

Key lesson: AI capability ≠ AI applicability. Barn architecture + breed are the applicability variables.

Sources: Wikimedia Commons (Holstein); Cargill press release 2018

L3 in Russia — Connectome.ai + sanction uncertainty of the dairy AI stack

Works — Connectome.ai

Skolkovo resident

*Narrow computer-vision task:
calving monitoring of calves*

+ early alert for assistance

*Pattern: narrow niche + CV +
measurable animal safety*

In question — DeLaval / GEA / Lely

Gray status of cloud analytics

- Equipment physically with farmers
- Cloud analytics depends on Europe
- Precedent: Climate FieldView 2022 → Microsoft Azure 2024 (for defense sector)
- Lobnya 2026 — equipment substitution started (₽4B), AI stack — separate track

F9 — "vapor" risk: the vendor can switch off AI functionality by sanction or license decision, without notifying users in advance.

Sources: Connectome.ai (Skolkovo); BUSINESS-stat 2026 Lobnya · TAdviser FieldView

4

Section 4 — L4 «Supply chain»

Agentic AI leads. Outcome is measured in basis points over minutes, not seasons.

4 working cases (Cargill CMAX, Tract, Olam, Walmart×Cropin) · 2 failures (USDA, Verra) · Russia parallel (X5/Magnit/RSHB)

Cargill CMAX — BIG AI Excellence Award 2026

Grain port and silos · Tilbury (Wikimedia, CC-BY-SA)

AI optimizes port forecasting and shipping logistics

70+ countries

of 195 (≈36% of the world) · ~155k staff

1000+ sites

warehouses / ports / silos of Cargill

BIG AI 2026

Excellence Award at the L4 level

Success principle: agent narrowness

One action per agent (hedge / route) + human-in-the-loop for trades >$10M notional.

Source: Cargill press release, April 2026 (BIG AI Excellence Award)

How the agent hedges — simplified 4-step flow

1. Sensor

State vector:

price + weather + currency

input data

2. Inference

Price distribution

5/30/90 days + uncertainty

ML model

3. Decision

4 actions:
buy / sell /
hold / hedge

+ human at >$10M notional

Action layer

4. Feedback

bp (basis points) over minutes,

online learning

Loop closes

Example: slippage cut from 45 bp to 8 bp

On a $5M notional trade: (45–8) × 0.01% × $5M = $1,850 per trade. At 20 trades per month = ~$37k saved. ROI of the narrow agentic AI — 1-2 months.

This works because the task is narrow (hedge timing), feedback is fast, and the human is only for large trades.

Example: McKinsey 2025 hedging report — typical manual slippage

Tract + Olam Mindsprint + Walmart × Cropin + Tesco

Tract

€18.6M Series A

Icos Capital · 4 anchor investors: Cargill + ADM + Olam + LDC.

Data backbone, NOT an agent.

Olam Procuresprint

Agentic procurement

Narrow procurement task: price + suppliers + contracts as an agent.

Walmart × Cropin

US + South Africa

(not India) — supply-chain analytics and ESG reporting.

Tesco AI

−30% food waste

Since 2017 — Tesco AI for perishables forecasting.

⚠ Don't confuse: Tract = data backbone (compliance infrastructure), not agentic AI.

Competitors jointly build compliance infrastructure — a rare Cargill+ADM+Olam+LDC collaboration precedent.

Sources: Tract press release 2024-08; FreshPlaza; Wipro press release 2026 (Mindsprint)

USDA Climate-Smart canceled + 94% phantom credits at Verra

April 2025 — USDA canceled

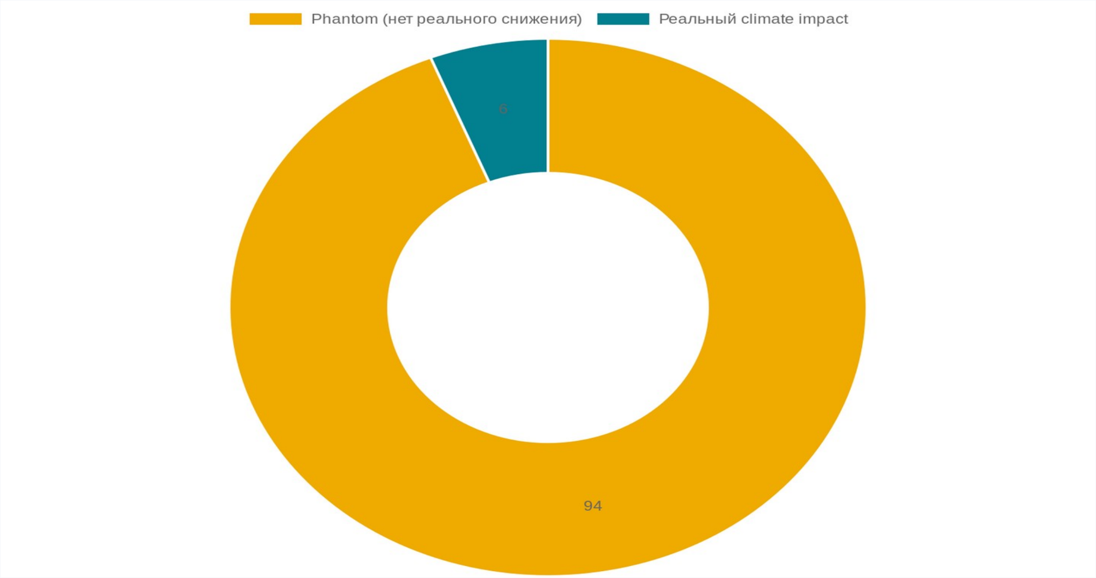
the Climate-Smart Commodities program

$3.1B	*funding · ≈$23M / project avg*
135 projects	*canceled*
14,000 farms	*lost support*
3.2M acres	*≈0.36% of 900M acres US ag*

Federal policy = tail risk
for business models dependent on subsidies

January 2023 — 94% phantom at Verra

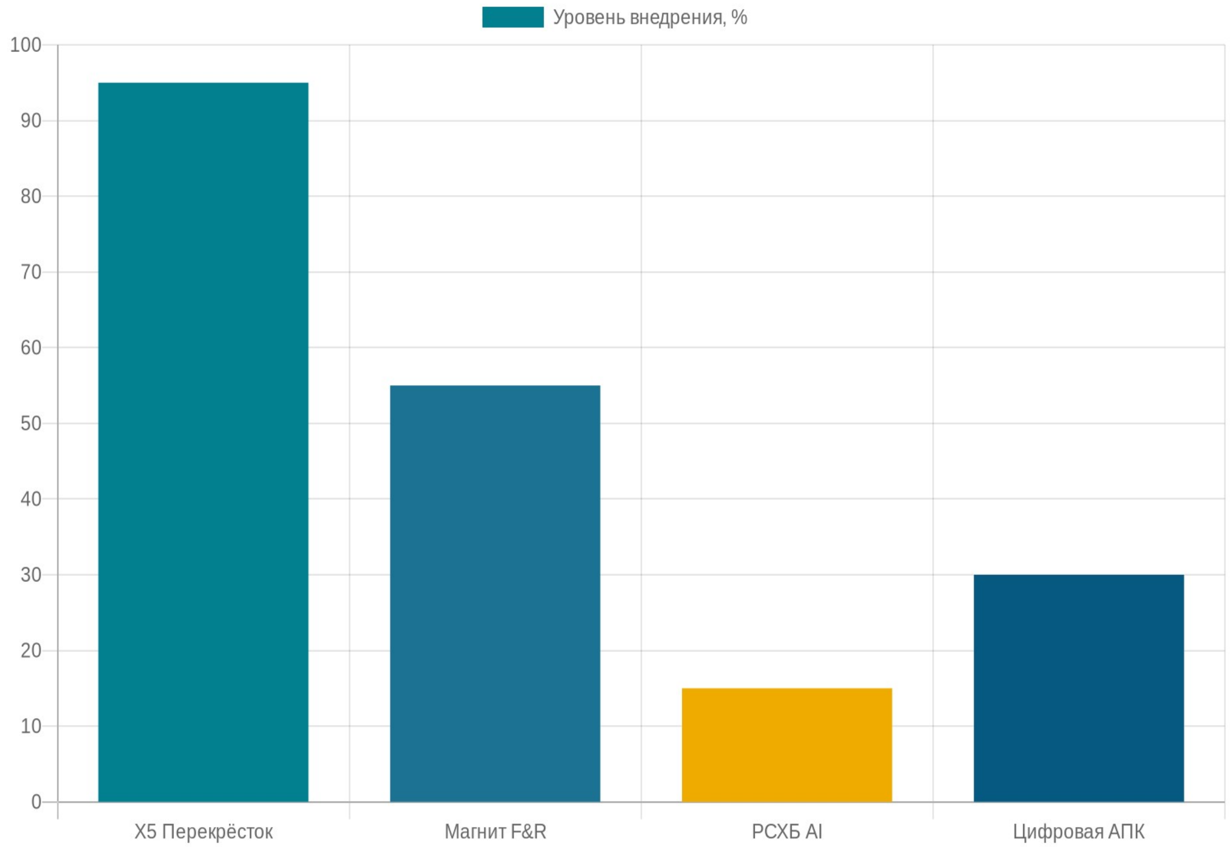
The Guardian + Die Zeit + SourceMaterial



94% of Verra forest credits → no real CO_2 reduction.
AP7: AI-MRV without direct measurement = scaled greenwashing.

Sources: USDA press 2025-04-14; The Guardian 2023-01; SourceMaterial 2023

L4 in Russia — X5 parity, Magnit hybrid, RSHB — "vapor"



X5 «Perekrestok»

ML since 2020 · 200+ factors

World-class. Works.

Magnit F&R

Hybrid: forecasting ✓, replenishment — pilot

Forecasting on 46 DCs Jan 2026 — in production.

Replenishment on 3 DCs — pilot.

RSHB AI services

Announced, no metrics

"Vapor" risk: declarations without production metrics.

Pattern: L4-L5 parity alongside an L1-L2 lag. GigaChat — a demo episode, not deployment.

Sources: Habr Magnit 2026-01; TAdviser X5 Tech 2024

4

Section 4-bis — Environment

Cross-cutting themes: connectivity · vendor lock-in · regulation. Without them, no rung of the ladder works.

3 sub-blocks (connectivity / vendor lock-in double optic / regulation) · 3 anti-AI criteria (AP5, AP6, AP7)

Connectivity — 18% of farms without internet + GNSS interference in Q1 2025

18%

≈360k of 2M US farms without internet (vs <1% urban)

123,000

≈1.2% of 10M flights Q1 2025 (vs <1k Q1 2022)

Apr 2026

Starlink ban in Russia

AP5 + alternative

AP5: a cloud-first approach for no-network settings = an architectural error.

Alternative: on-device ML (edge / TinyML)

- *model inference right on the device*
- *minimal cloud connectivity*
- *works under GNSS interference*

Sources: Stanford GPS Lab ITM 2025; ICAO 2025 (Switzerland / Finland / Estonia)

Vendor lock-in — the John Deere double optic

May 2022 — anti-theft success

Melitopol, occupied territory

*Remote John Deere lock-out:
27 units of machinery
worth $5M turned into
bricks by the manufacturer's decision.*

AI security feature — a success within this loop

Jan 2025 — FTC v. Deere

Decade-long repair restrictions

*The same mechanism:
remote lock-out = security today =
a control surface tomorrow. The farmer
cannot repair without
the vendor's permission.*

December 2025 — FCC banned DJI

*80-90% of US agricultural drones lost
legal status. One FCC decision —
vendor lock-in became a geopolitical risk.*

Sources: The Register 2022-05-02; CSO Online 572811; FTC press release 2025-01-15

Regulation — EU AI Act vs USDA vs "Agriculture of the Future"

EU AI Act

Regulation 2024/1689

- Ag machinery — high risk
- Feb 2025 — operator obligated
- AI literacy mandatory
- Liability chain: vendor → operator

Strict, enforceable

USDA AI Strategy

FY2025-26

- Formal declaration
- Climate-Smart canceled
- No mandatory AI literacy
- No high-risk classification

Formal, weak enforcement

Russia "Agriculture of the Future"

Decree 31.12.2025

- Declarative program
- Previous one (Agriculture 2024) –3.2%
- No measurable indicators
- No vendor liability

Declarative, unverified

Sources: Cambridge EJRR 2024; USDA AI Strategy FY2025-26; Decree of the Government of the Russian Federation 31.12.2025

5

Section 5 — L5 «Shelf» + 5 criteria

Retail AI is mature + five anti-AI criteria + checklist + career landscape + callback to Plenty.

L5 briefly (Walmart, Tesco, X5) + 5 criteria + checklist + careers + closing callback

L5 — Walmart Eden + Tesco AI + X5 «Perekrestok» + Magnit F&R (hybrid)

Walmart Eden

ML since 2017

Perishables forecasting +
freshness routing across 11,000+ stores

Tesco AI

−30% food waste

Since 2017 — daily forecasting
for ~3,500 UK supermarkets

X5 «Perekrestok»

ML since 2020 · 200+ factors

Russian L5 flagship.
World-class.

Magnit F&R (hybrid)

Forecasting ✓ · Replenishment — pilot

Forecasting on 46 DCs January 2026 — in production.
Replenishment — pilot on 3 DCs.

⚠ *Caveat: L5 is retail AI, not agriculture. Magnit F&R — the "hybrid" nuance: forecasting in production, replenishment a pilot.*

Walmart corporate 2025 · Tesco AI annual report · X5 Tech 2024 · Habr Magnit 2026-01

Five anti-AI criteria — the key takeaway of the lecture

#	Criterion "when not AI"	Failure example	Alternative
AP1	**Thermodynamics > AI**	*Plenty Compton: $940M in losses*	**Not AI · open field or different unit economics**
AP3	**Accuracy ≠ deployment**	*Plantix 85% × 10M = ~100k errors*	**Calibrated confidence + abstention**
AP4	**Generic LLM as advisor**	*Tzachor: 56-71% wrong*	**AI with source verification + human-in-the-loop**
AP6	**AI equipment = vendor lock-in**	*Deere $5M + FTC + DJI ban*	**Open standards / multi-vendor**
AP7	**AI-MRV without direct measurement**	*Verra: 94% phantom credits*	**Soil sampling + satellite physics**

Also: AP2a (architecture choice within AI) · AP2b (deterministic alternative) · AP5 (cloud-first off-grid)

Pre-purchase verification checklist for an AI solution

Block 1

Task classification

1. Narrow or general-purpose?
2. Closed-loop or open environment?

Block 2

Production status

3. Production metrics vs announcements?
4. Is there independent verification?

Block 3

Accountability

5. Vendor liability chain?
6. SLA on AI-functionality failures?

Block 4

Vendor lock-in

7. Open data standards?
8. Interoperability with other vendors?

Block 5

Connectivity

9. On-device fallback when internet is absent?
10. Alternative to GNSS navigation?

Scoring: 8-10 "yes" = buy/pilot · 5-7 = conditional · ≤4 = reject

Answer — not in the vendor's words, but from external sources: independent audits, FTC press releases, industry reports.

Career landscape — L1-L5 segments + the Russian market

L1 Field

International

John Deere · Bayer · BASF
Climate · Taranis · Granular

Russian

Cognitive Pilot · ITELMA
EFKO · Geoscan · ExactFarming

L2 Robot

International

Carbon Robotics · Saga · Tevel
AGCO · Aigen

Russian

(niche RU startups)

L3 Animal

International

DeLaval · GEA · Lely
Cargill Birdoo · Allflex

Russian

Connectome.ai · Skolkovo

L4 Supply chain

International

Cargill CMAX · Tract · Olam
ADM · Walmart×Cropin

Russian

X5 Tech · Magnit digital
Rusagro Tech · RSHB.digital

L5 Shelf

International

Walmart Eden · Tesco AI
WooliesX · Carrefour AI

Russian

X5 «Perekrestok»
(world-class)

Market landscape without advocacy. Generic form — "specialized technical universities"

Callback to the start of the lecture (Plenty Compton):

Plenty did not close because of bad AI.

Plenty closed because of LED thermodynamics.

The controller worked. CV recognized. The model trained. LED ≈ 100× sunlight energy — AP1.

For each rung of the ladder — what works and what breaks:

Level	✓ Works	✗ Breaks
L1 Field	*See & Spray (5M acres)*	*Plenty / Bowery (thermodynamics)*
L2 Robot	*LaserWeeder G2 (narrow laser)*	*Monarch (demo ≠ deployment)*
L3 Animal	*SenseHub (2M cows)*	*Cainthus / tie-stall housing*
L4 Supply chain	*Cargill CMAX (narrow agent)*	*USDA Climate-Smart / Verra 94%*
L5 Shelf	*Walmart Eden / Tesco / X5*	*GNSS jamming Finland*

The engineer holds the whole ladder in mind

and picks the right tool for each rung.

Knows where AI does not work — and why.

Transition to Lecture 11

L11 — cyber-physical manufacturing. A closed loop like L4-L5 + physical contact of AI with the product like L2.

Thank you. Next — questions and answers

Questions and answers

Backup questions on VF thermodynamics, agentic AI scope, ITELMA vs Cognitive Pilot, foundation models

Backup questions

Plenty and Bowery

Why vertical farms specifically in the US?
In a closed environment — US vs Japan vs UAE?

Agentic AI

Where else does agentic AI work
in engineering tasks — besides hedging?

Cognitive and ITELMA

Can a hybrid solution be built
CV + sensor fusion, or are these different paradigms?